

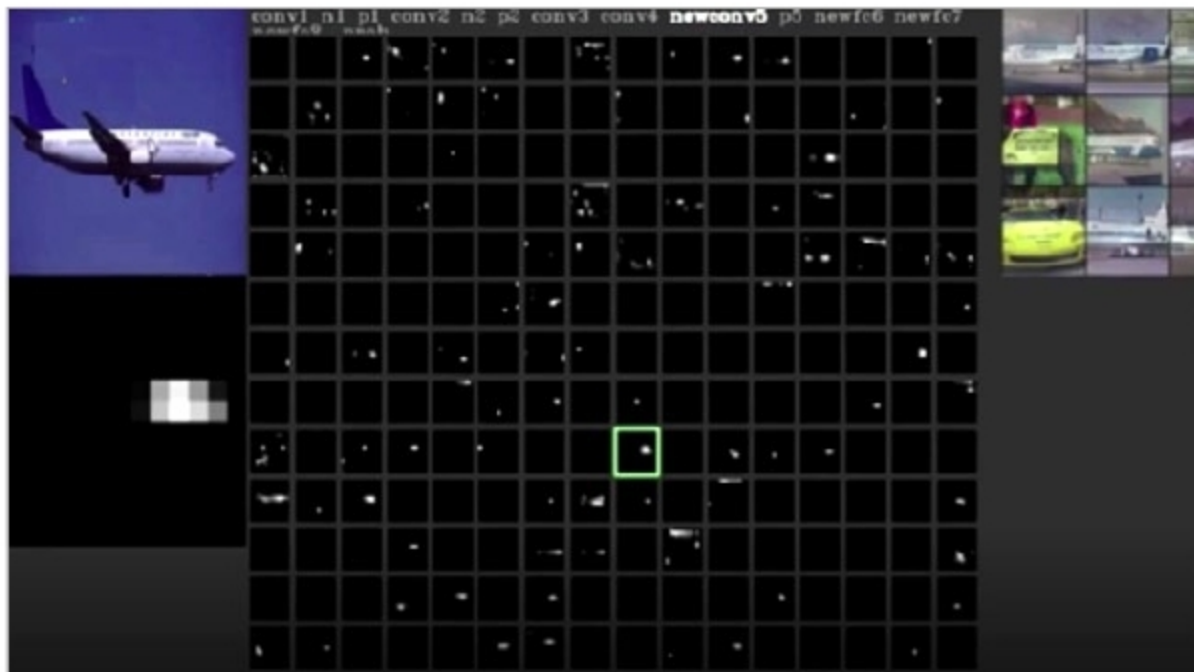


Figure 4: Activation feature maps of a passenger flight

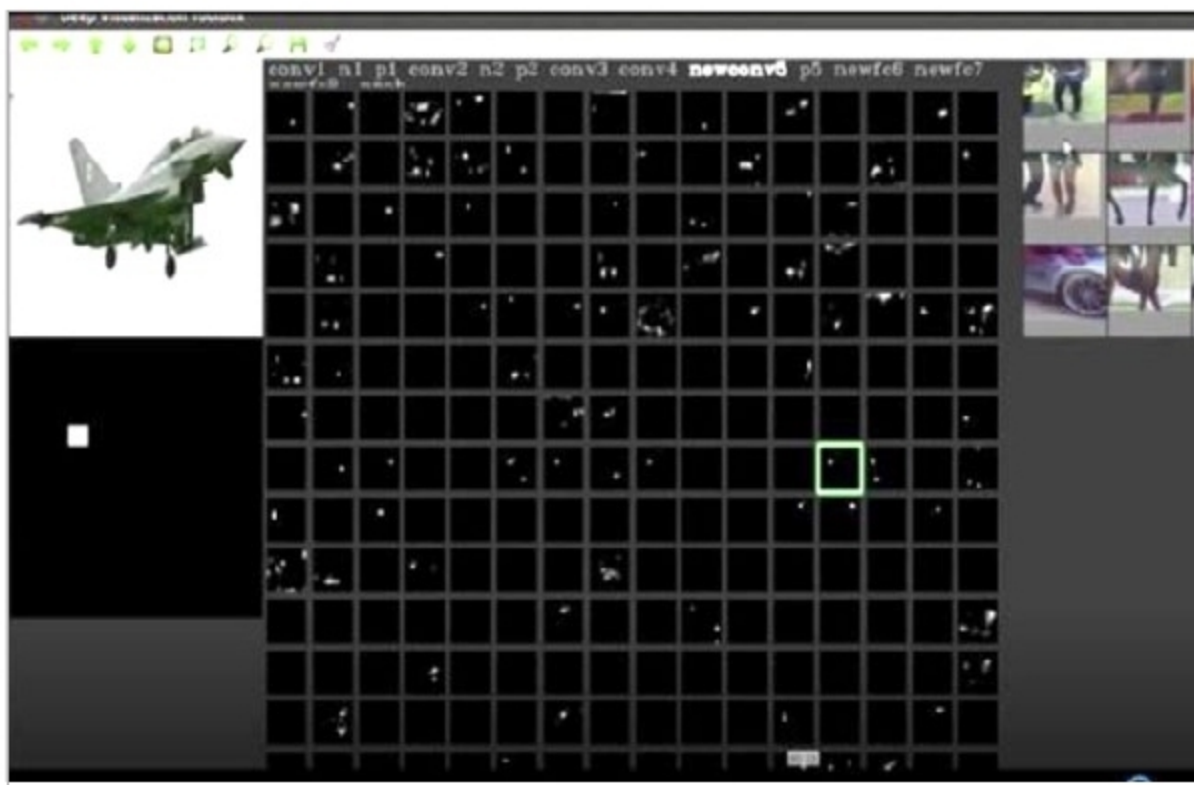


Figure 5: Activation feature maps of fighter flight

layers — the first four convolutional layers are frozen and the model is trained to obtain very good accuracy of around 98 per cent. The figures 4 and 5 show the visualizations of the activation feature maps of the fifth convolutional layer, which has 256 filters and hence 256 activation feature maps. Figures 4 and 5 show the following details.

- The activation feature maps for the passenger flight image and the fighter flight image for the newconv5 convolution layer (the fifth convolutional layer of the AlexNet model, renamed to newconv5 since it has been

retrained during transfer learning).

- The passenger and fighter flight test images (top left corner in the figure) for which visualisations are shown.
- Assume the activation feature maps are arranged as a 16 x 16 matrix with index 1 given for activation feature map at row 1, column 1; and index 17 for the activation feature map at row 2, column 1, and so on.
- The higher resolution image of the selected activation feature map (green coloured box) is shown just below the test image.

In Figure 4, check the 137th

activation feature map (the green coloured box), which is shown as a high resolution image below the test image. The white coloured pixels in this image show only those features that have been learnt by this filter or, in other words, these are the activated neurons of the feature map with a value greater than zero. The remaining pixel values are zero and hence black in colour, so neurons corresponding to those pixels are not activated. From a comparison between the activation feature map and the original image, it can be seen that the features in the activation feature map correspond to multiple windows in the input image. This can also be justified by checking the top nine highly activated images shown at the top right corner in the figure. The top nine images show the objects that have multiple window patterns or features. Refer to <http://yosinski.com/deepvis> for more details on visualisation of the CNN model. A similar analogy is applicable to neurons in all the feature maps shown above.

Now, among the 16 x 16 activation feature maps in the convolution layer shown in Figures 4 and 5, it can be seen that many of these maps are not showing any activations. None of the neurons in these feature maps are activated. CNN theory states that each filter represents distinct feature/s at each layer, and in these figures, each of the 256 filters represents features of the passenger or the fighter flight that are learnt. If there are no activations, this means that it does not learn any feature. Since the model is a binary classifier for classifying passenger and fighter flights, each of the filters in the convolution layer will be learning either passenger or fighter features. Hence these feature maps can be classified into four types – class 0, class 1, mixed and inactive.

- **Class 0:** In these feature maps, the